SATHISH KUMAR N

EMBEDDED DEVELOPER

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PROFESSIONAL SUMMARY

To build a rewarding career in the Embedded Systems domain by applying my passion for Electronics, Microcontrollers and C/C++ programming to create innovative and efficient solutions, while growing alongside a forward-thinking organisation and making a meaningful technical contribution.

TECHNICAL SKILLS

Programming Language:	C Programming, Embedded C, C++ Programming
Hardware Knowledge:	Circuit Debugging, Basics of Electronics, Soldering, PCB Design
Microcontrollers:	8051, PIC16F877A, LPC2148, STM32
MCU Concepts:	Registers, GPIO, ADC, Interrupts, EEPROM, Communication Protocols
Tools:	MPLAB XIDE, Keil, STM32 CubeMx, Proteus, EasyEDA, Altium

SOFT SKILLS

Professional Skill:	Project Management, Team Collaboration, Adaptability, Critical Thinking
Language:	Tamil(R/W/S), English(R/W/S)

PROFESSIONAL EXPERIENCE

- Embedded System Internship (6 months) - PUMO Technovation
- Medical Equipment Hands-on Workshop (10 days) - KAV Biomedical Equipment Training and Education
- Hospital Internship Training (10 days) - OMS Hospital, Tiruppur

EDUCATION BACKGROUND

S.No.	Institution Name	Qualification	Score	Duration
1.	Velalar College of Engineering and Technology	B.E. Medical Electronics	7.12-CGPA	2022 - 2026
2.	Navarasam Matriculation Higher Secondary School	HSC	69 %	2021 - 2022
3.	Navarasam Matriculation Higher Secondary School	SSLC	95%	2019 - 2020

PROJECT

Project Title: Non -Invasive Blood Vessel Block Detection

Presented this Innovative Project in Smart India Hackathon (SIH) Grand Finale 2024 under the Medical and Healthcare Theme. The System uses Doppler Ultrasound, AI-based Imaging, and Signal Processing to detect Blockages without Radiation. Includes ECG, Blood Pressure Monitoring for Accurate results.

Project Title: Innovative Intravenous Techniques

Developed a Smart IV Saline Monitoring System using a Non-contact liquid level detection sensor and GSM- based IoT alerts for low Saline levels. Integrated a servo motor to prevent backflow and air embolism by controlling flow automatically. Included a patch-type Secondary container for Uninterrupted flow and a nurse call button for Emergency response. Presented in the College of M.P.Nachimuthu M.Jaganathan Engineering College secured the 3rd Prize.

ACHIEVEMENTS

S.No.	Project Title	Participation	Winnings
1.	Non-Invasive Blood Vessel Block Detection	Shree Venkateshwara Hi-Tech Engineering College	**1st** Prize
2.	Smart Neck Support Band	Inter College Project Expo	**2nd** Prize
3.	Smart IV Saline Monitoring System	Adithya Institute of Technology	**3rd** Prize